Fundamentals of Software Engineering, Individual Assignment [20%]

Due Date: 16 Dec 2025, 11:30 AM

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I.

For this assignment, you are asked to investigate how a software-related idea or concept can be developed into a viable business opportunity. Begin by identifying a problem, need, or opportunity that could be addressed through software. How can you explore and evaluate different ways of solving this problem, considering both technical feasibility and real-world impact? What criteria would you use to determine which solution is most promising, and how would you test or validate your assumptions about its usefulness or value?

Next, how might this solution be shaped into a business concept? Consider questions such as: Who would be the target users or customers? What value would your solution provide, and why would people or organizations be willing to adopt or pay for it? How can you prioritize features, functionality, or services to maximize its potential success in the market? What are the possible challenges, risks, or limitations in bringing your idea to life as a software product or service?

Finally, reflect on the process of moving from an idea to a business-ready software solution. What insights did you gain about how innovative thinking, problem-solving, and understanding user needs interact to create value? How can software engineers contribute not only to technical solutions but also to generating sustainable business opportunities?

II.

For this assignment, you are asked to explore your interest and aptitude in creating new technologies, algorithms, or software solutions that address real-world challenges or opportunities. Begin by identifying a problem, process, or area where you see potential for innovation; this could be in areas such as data processing, automation, optimization, artificial intelligence, user interaction, or system efficiency. How would you approach designing a new algorithm, software tool, or technological solution to address this problem, and what guiding principles would you follow in your design? Consider how you might evaluate the effectiveness, efficiency, or novelty of your solution, and what challenges you might anticipate during development. As you reflect on this process, what excites you about creating new technology, and how do your ideas reveal your potential talent, creativity, and future interests in computer and software technologies? Finally, discuss how exploring such problems and attempting to design innovative solutions can help you understand your own strengths and areas for growth as an aspiring software engineer.

Assignment Submission Should not Exceed Three Pages Including the Cover Page